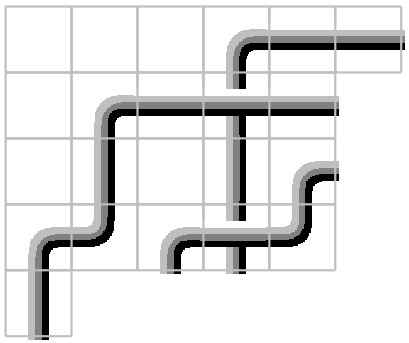
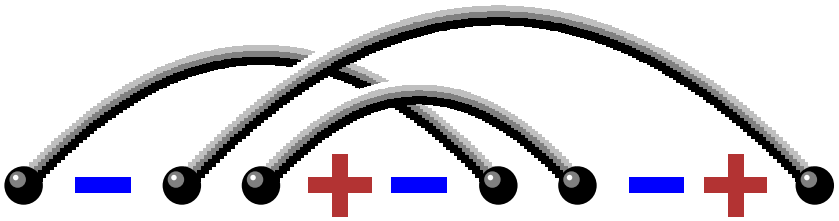
Bumpless Pipe Dream Fragments —Equivariant Geometry of Clans

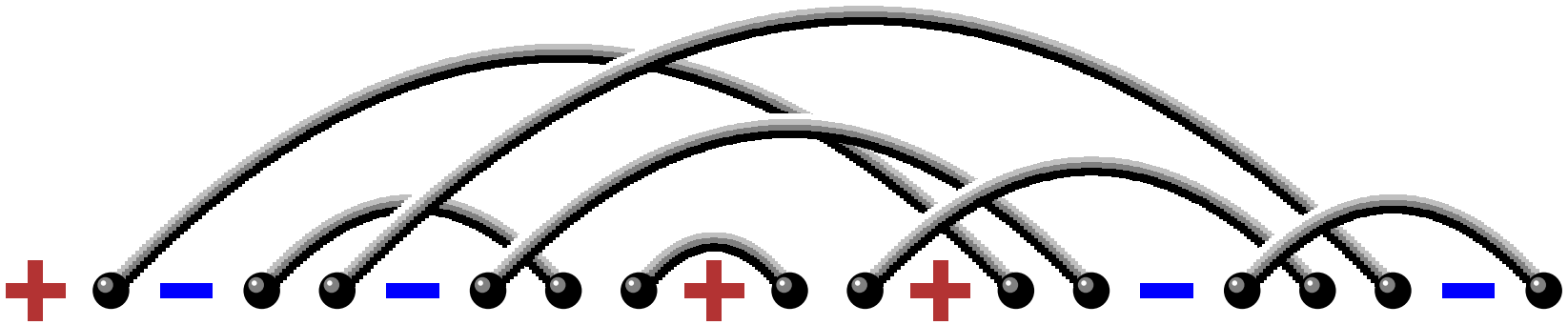
(Joint with Yiming Chen, Neil J.Y. Fan and Ming Yao)

Rui Xiong





Combinatorics of Clans



Symmetric Pairs

A **symmetric pair** is a pair (K, G) where $K = G^\theta$ for some involution θ of G. When $G = GL_n$, we have three types

type	involution	$K \subset G$
AI	$g \mapsto (g^t)^{-1}$	$O_n \subset GL_n$
AII	$g \mapsto J_{2m}(g^t)^{-1}J_{2m}$	$Sp_{2m} \subset GL_{2m}$
AIII	$g \mapsto I_{p,q} \cdot g \cdot I_{p,q}^{-1}$	$GL_p \times GL_q \subset GL_{p+q}$

where $J_{2m} = \begin{bmatrix} & I_m \\ -I_m & \end{bmatrix}$ and $I_{p,q} = \begin{bmatrix} I_p & \\ & -I_q \end{bmatrix}$.

Real Groups

Symmetric pair arises from the study of real groups.

type	$K \subset G$	real form
AI	$O_n \subset GL_n$	$GL_n(\mathbb{R})$
AII	$Sp_{2m} \subset GL_{2m}$	$GL_m(\mathbb{H})$
AIII	$GL_p \times GL_q \subset GL_{p+q}$	$U(p,q)$

This will allow us translate real question to complex questions. For example, the geometry of K-orbit closure over flag variety is closely related to the representation of real group, thanks to the serial work of Vogan.

Spherical Subgroups

A subgroup H of G is called **spherical** if the set

$$H\backslash G/B = \{H\text{-orbits of } gB \in G/B\}$$

is finite, where G/B is the flag variety.

The Borel subgroup B is spherical. We have **Bruhat decomposition**

$$B\backslash G/B \overset{1:1}{=\!=} \text{Weyl group.}$$

Each orbit is known as a **Schubert cell**.

The group K in a symmetric pair is spherical (a version of **Iwasawa decomposition**).

$$K\backslash G/B \overset{1:1}{=\!=} \text{a finite set.}$$

We are interested in the K-orbits of G/B.

Index set of K-orbits

When $G = GL_n$, the flag variety

$$G/B = \{0 = V_0 \subset V_1 \subset \cdots \subset V_{n-1} \subset V_n = \mathbb{C}^n\}.$$

type	$K \subset G$	index set of $K \backslash G/B$
AI	$O_n \subset GL_n$	$\{w \in S_n : w^2 = 1\}$
AII	$Sp_{2m} \subset GL_{2m}$	$\{w \in S_{2m} : w^2 = 1, w(i) \neq i\}$
AIII	$GL_p \times GL_q \subset GL_{p+q}$	$\{(p, q)\text{-clans}\}$

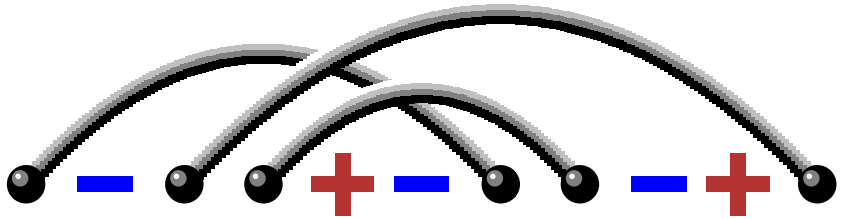
Today we will focus on type AIII.

Clans

A (p, q)-**clan** is a partial matching of $p + q$ nodes with unmatched nodes colored by $+$ or $-$ with

$$\#\{+\} - \#\{-\} = p - q.$$

For example, the following is a $(5, 6)$-clan

$\gamma =$ 

$$\boldsymbol{v}_i^{\gamma} = \overset{e_1}{\underset{e_8}{+}}\ e_6\ \overset{e_2}{\underset{e_{11}}{+}}\ \overset{e_3}{\underset{e_9}{+}}\ e_4\ e_7\ e_8\ e_9\ e_{10}\ e_5\ e_{11},$$

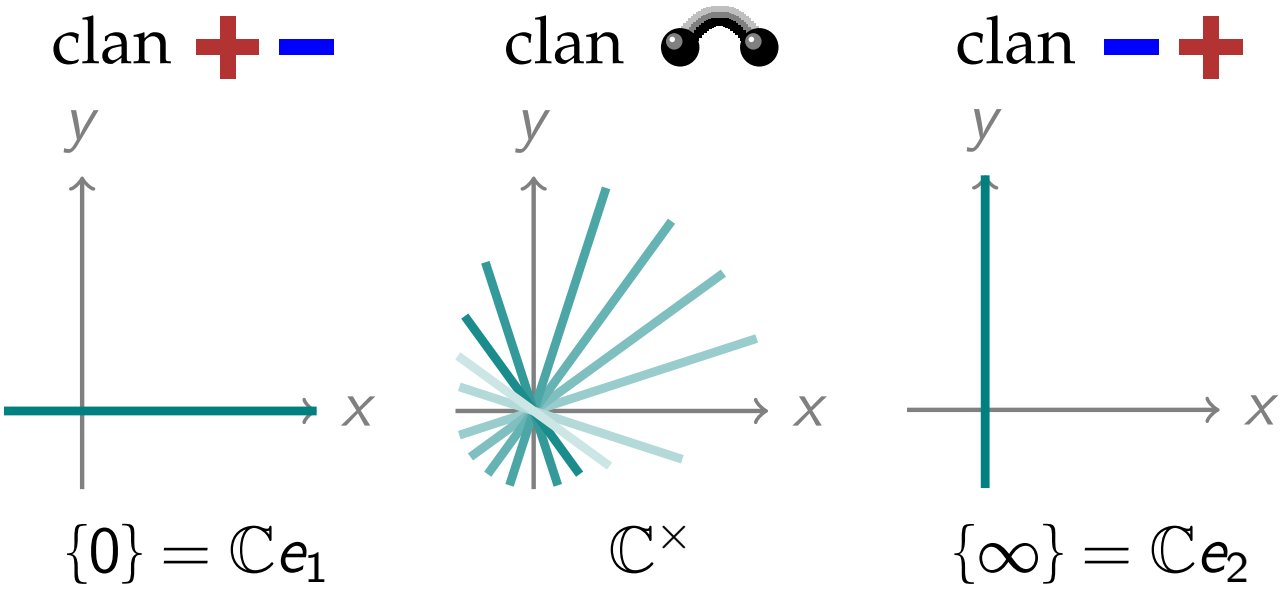
$$\text{the orbit} = K \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} B/B.$$

Example

Consider the case

$$GL_1 \times GL_1 \subset GL_2 \circlearrowright \mathbb{P}^1 = \{0 \subset L \subset \mathbb{C}^2\} = \mathbb{C} \sqcup \{\infty\}.$$

There are three orbits

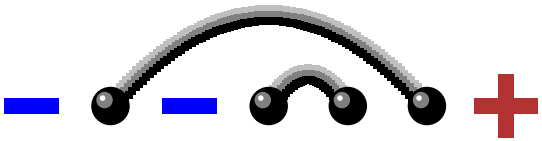


Special types of clans

matchless

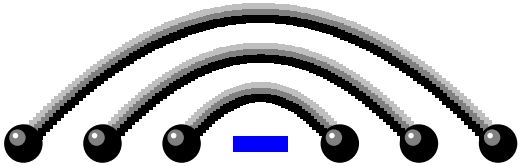
closed K-orbits
(there are $\binom{p+q}{p}$ many)

non-crossing



Richardson varieties (indexed by two inverse Grassmannian permutations)

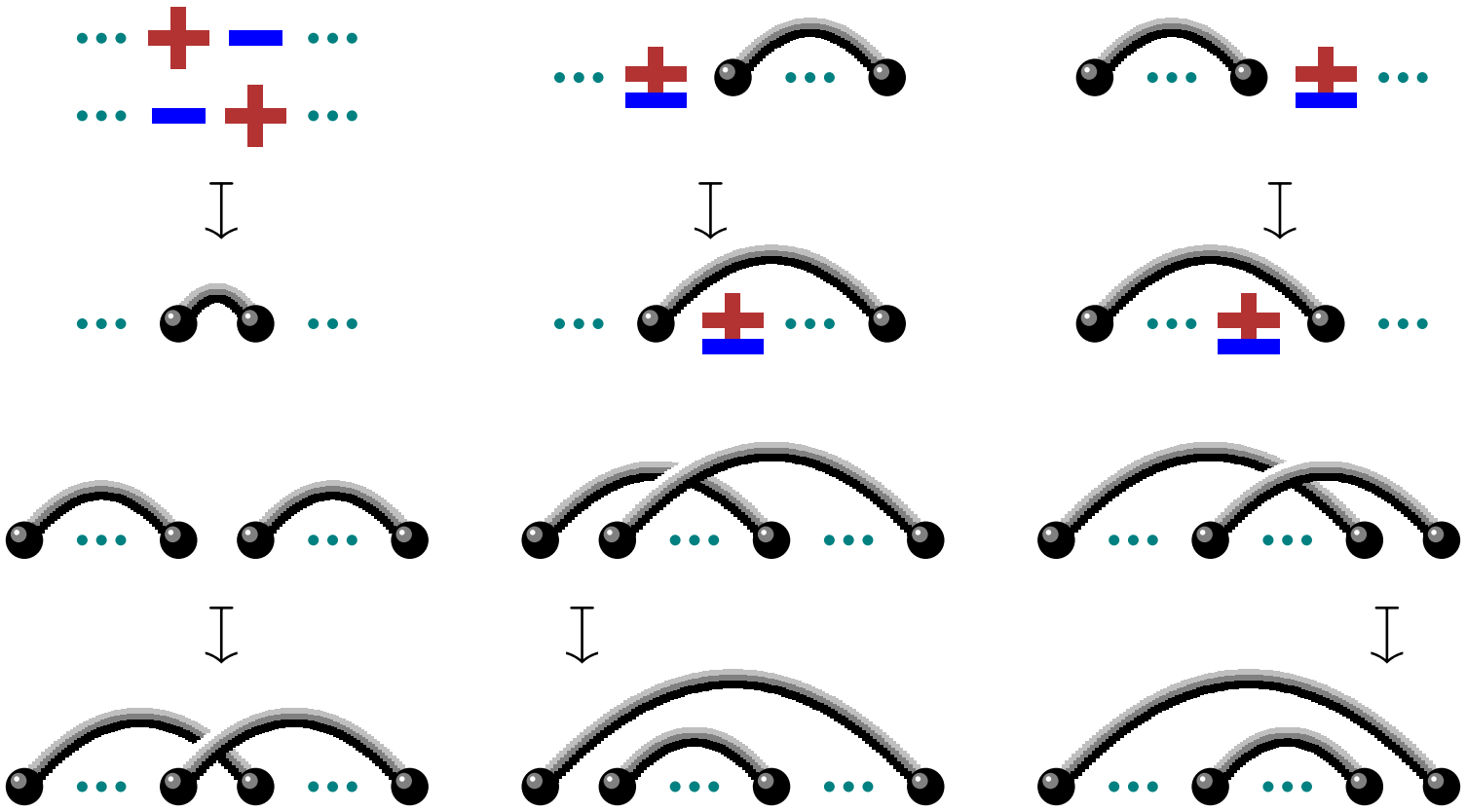
rainbow



the open orbit
(open orbit is unique)

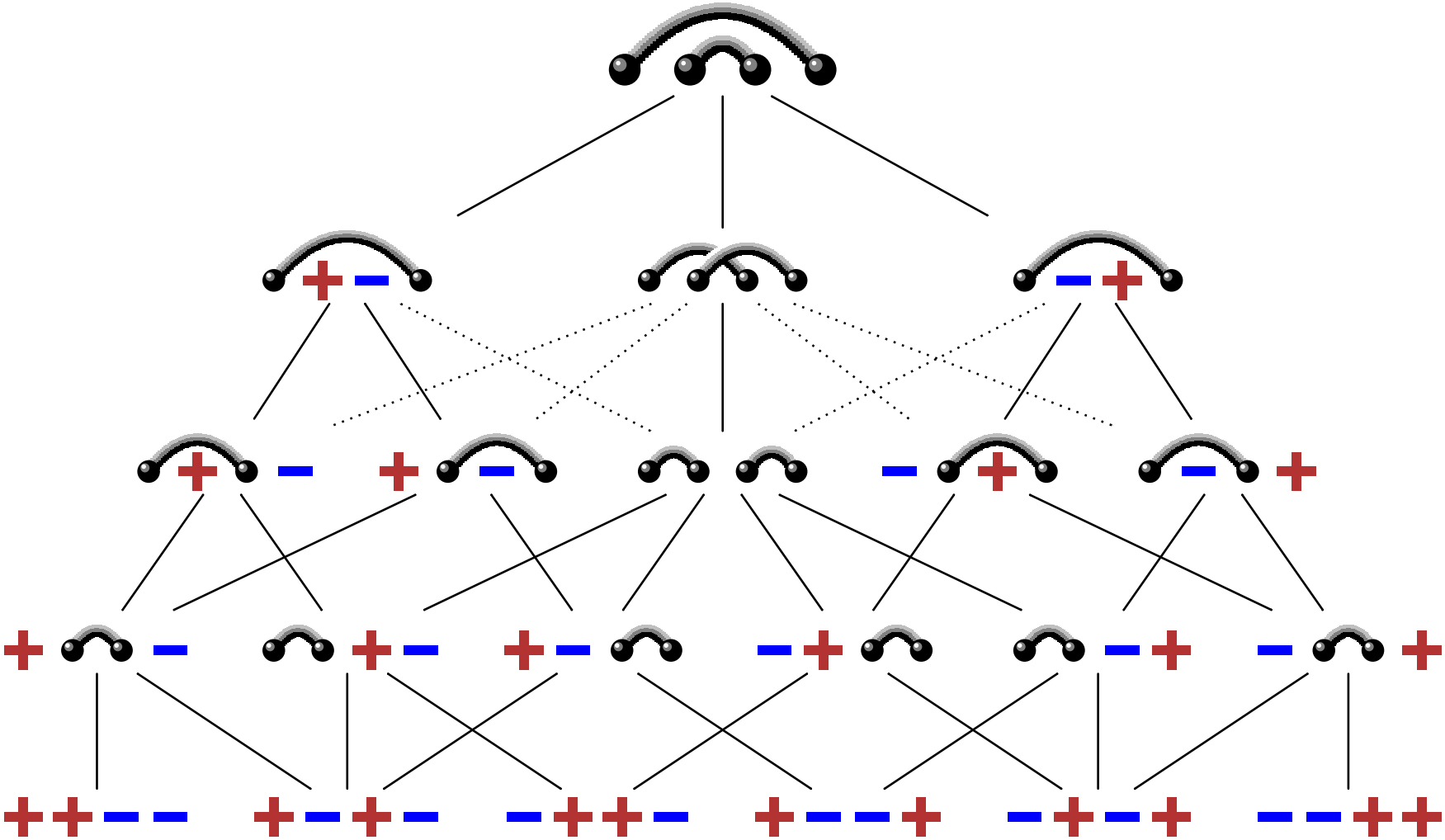
Orders on Clans

The weak order of clans are generated by



There is also a strong order induced by orbital closure.

Example



Schubert Calculus

In the work of Wyser and Yong [WY14], many fundamental framework was built.

- **polynomial representations**
- **non-equivariant Schubert expansion**
- **commutative algebra interpolations** and **etc.**

However, there remain some questions

- **combinatorial models of the polynomials**
- **equivariant Schubert expansion** and **etc.**

We will answer the second question.

B. J. Wyser and A. Yong. Polynomials for $GL_p \times GL_q$ orbit closures in the flag variety. Sel. Math. New Ser. (2014) 20:1083-1110.

Polynomial representatives

The ring $H_T^*(G/B)$ is a quotient of polynomial ring (in two family of variables). The fundamental question is to find a polynomial representative.

The classes of B-orbit closure are represented by **double Schubert polynomials** by Lascoux and Schützenberger [LS82].

The representative of the cohomology class of K-orbit closure is determined by Wyser and Yong [WY14].

- A. Lascoux and M. Schützenberger, Polynômes de Schubert, C. R. Math. Acad. Sci. Paris, Sér. I Math. 294(1982), 447–450.
- B. J. Wyser and A. Yong. Polynomials for $GL_p \times GL_q$ orbit closures in the flag variety. Sel. Math. New Ser. (2014) 20:1083-1110.

Definition

Recall the divided difference operator

$$\partial_i f = \frac{f - f|_{x_i \leftrightarrow x_{i+1}}}{x_i - x_{i+1}}.$$

Schubert polynomials are characterized by

$$\mathfrak{S}_{w_0} = \prod_{i+j \leq n} (x_i - y_j)$$

and

$$\partial_i \mathfrak{S}_w = \begin{cases} \mathfrak{S}_{ws_i}, & ws_i < w, \\ 0, & ws_i > w. \end{cases}$$

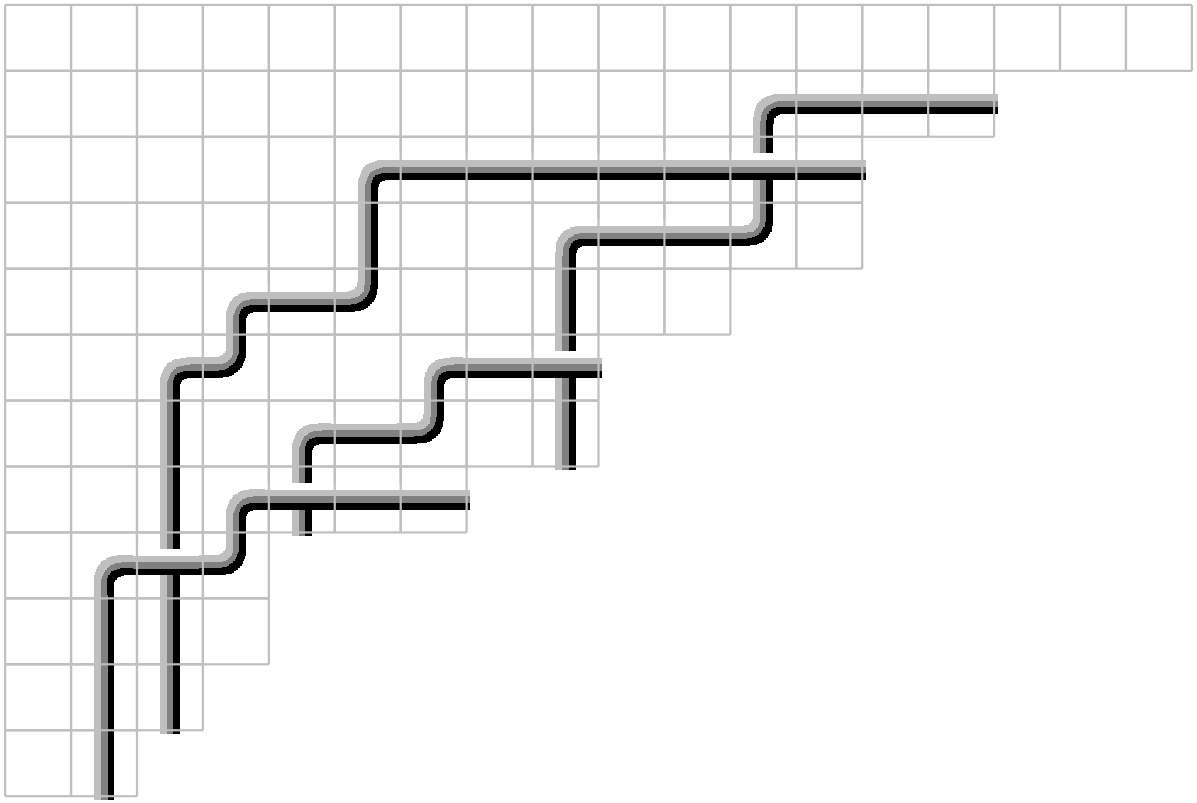
The polynomials of Wyser and Yong are characterized by

$$\Upsilon_\gamma(x, y) = \mathfrak{S}_{u_\gamma}(x, y) \cdot \mathfrak{S}_{v_\gamma}(x, \overleftarrow{y})$$

when γ is non-crossing, and

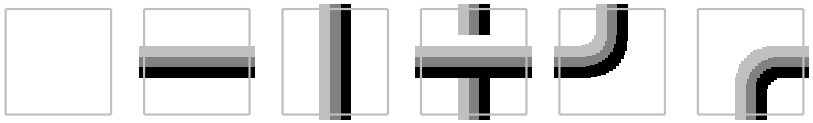
$$\partial_i \Upsilon_\gamma = \begin{cases} \Upsilon_{s_i * \gamma}, & s_i * \gamma \neq \gamma, \\ 0, & s_i * \gamma = \gamma. \end{cases}$$

Bumpless Pipe Dreams
Fragment

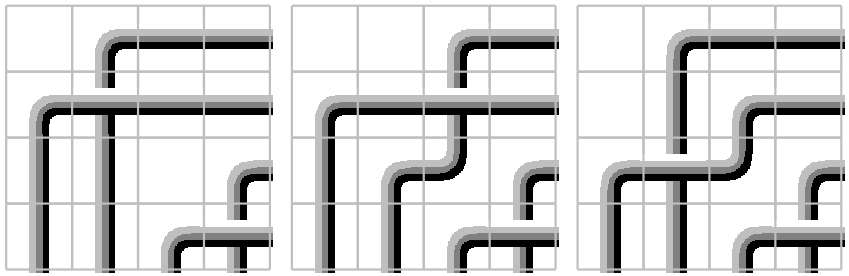


-less Pipe Dreams

A BPD of $w \in S_n$ is a tiling of $n \times n$ grid via



such that no pair of pipes intersects more than once BPDs can be used to compute double Schubert polynomials.

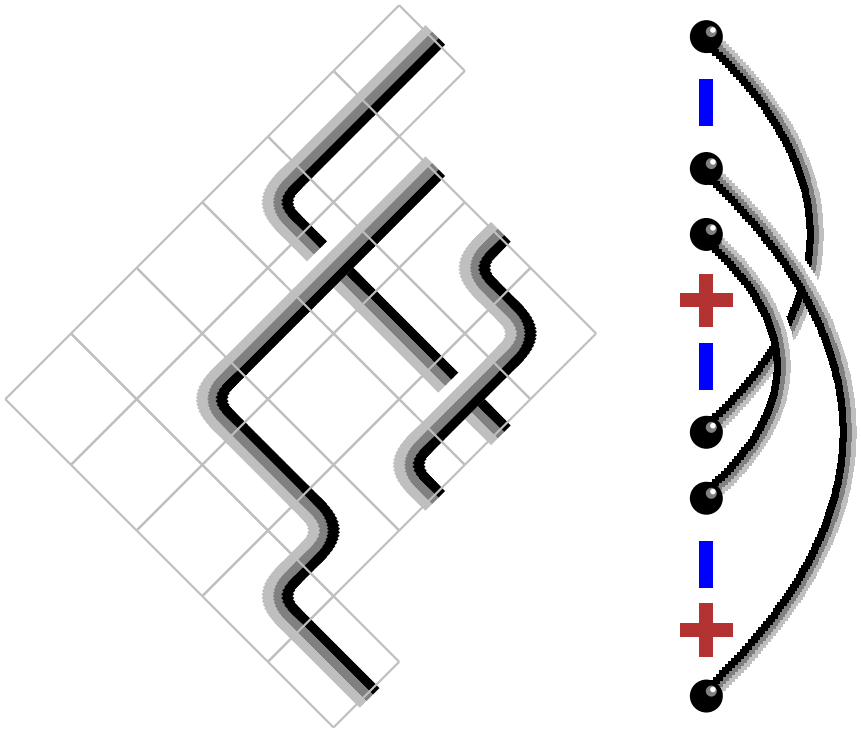


$$\begin{aligned}\mathfrak{S}_{2143} = &(x_1 - y_1)(x_3 - y_3)\\ &+ (x_1 - y_1)(x_1 - y_2)\\ &+ (x_1 - y_1)(x_2 - y_1).\end{aligned}$$

- T. Lam, S. J. Lee, and M. Shimozono. Back stable Schubert calculus. Compos. Math. **157** (2018), 883–962.
- A. Weigandt. Bumpless pipe dreams and alternating sign matrices. J. Combin. Theory Ser. A **182** (2021), 105470.

BPD fragments

BPDs fragments are BPDs supported on partitions, as considered by Weigandt [Wei].



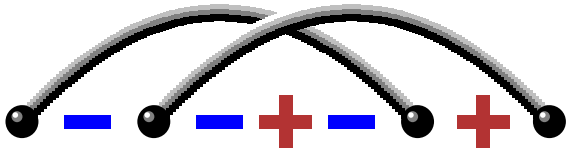
The novel observation is that BPD fragments are naturally indexed by **clans**, rather than permutations.

Clan polynomial

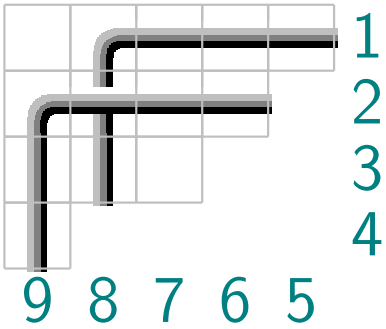
For a (p,q)-clan γ, we define the **clan polynomial**

$$\mathfrak{s}_\gamma(x) = \mathfrak{s}_\gamma(x_1,\ldots,x_{p+q}) := \sum_{\pi\in\mathrm{BPD}(\gamma)} \mathrm{wt}(\pi),$$

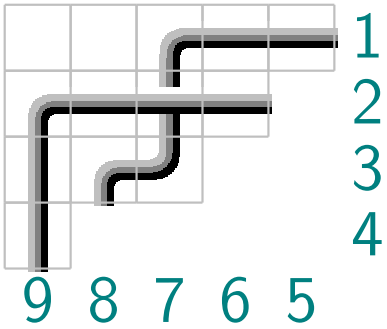
$$\text{where } \mathrm{wt}(\pi) = \prod_{(i,j) \text{ is an empty tile } \square} (x_i - x_{n-j+1}).$$

For example, let $\gamma =$ 

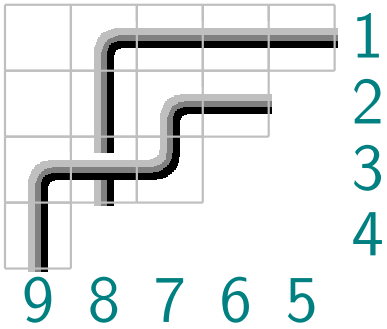
, then $\mathfrak{s}_\gamma(x)$



$(x_1 - x_9)(x_3 - x_7)$



$(x_1 - x_9)(x_1 - x_8)$



$(x_1 - x_9)(x_2 - x_9)$

Main theorem

Let us denote the localization

$$\Upsilon_\gamma|_w = \Upsilon_\gamma(y_{w(1)}, \ldots, y_{w(n)} \mid y_1, \ldots, y_n).$$

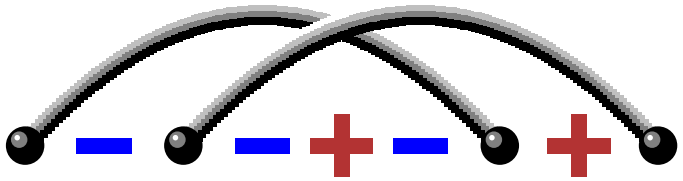
Our main result is a combinatorial formula of the localization at the minimal $w = v_\gamma$ such that $\Upsilon_\gamma|_w \neq 0$.

Theorem (Chen–Fan–X.–Yao, 2025+)

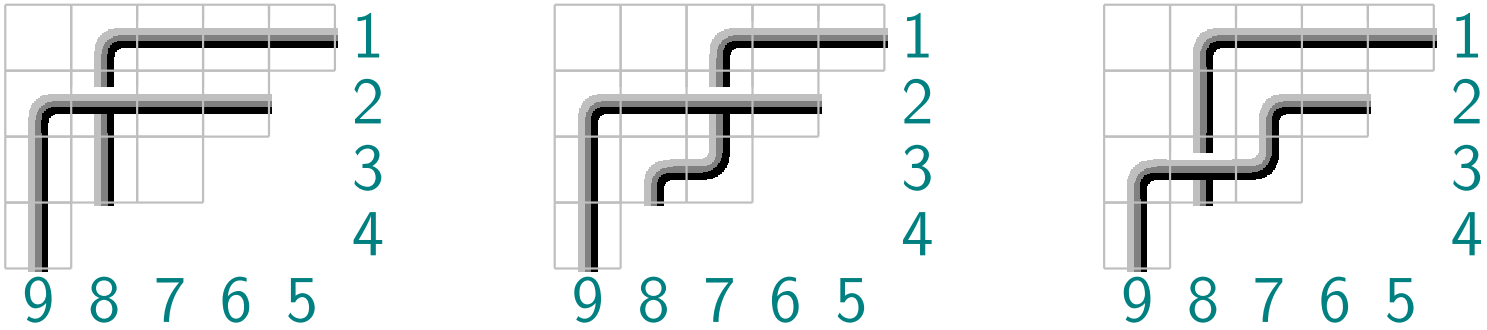
The localization $\Upsilon_\gamma|_{v_\gamma} \in \mathbb{Q}[y_1, \ldots, y_n]$ *is given by* **clan polynomials***:*

$$[Y_\gamma]_T|_{v_\gamma} = \Upsilon_\gamma(x; y)|_{v_\gamma} = \mathfrak{s}_\gamma(y) \cdot \prod_{(i,j) \in \overline{\lambda(\gamma)}} (y_{n-j+1} - y_i).$$

Example

$\gamma =$ 

$v_\gamma = 1 \quad 5 \quad 2 \quad 6 \quad 3 \quad 7 \quad 8 \quad 4 \quad 9$

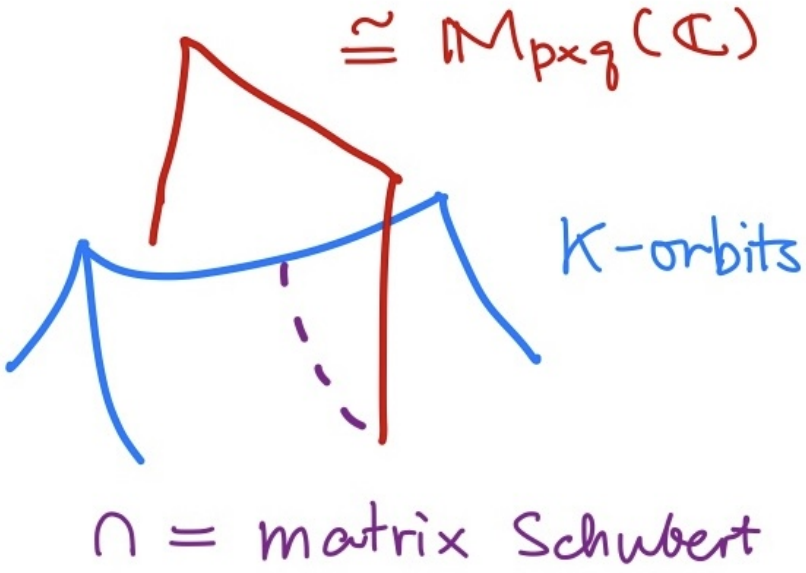


$$\begin{array}{ccccc} & & & & y_5\text{-}y_2 \\ & & & y_6\text{-}y_3 & y_5\text{-}y_3 \\ y_8\text{-}y_4 & y_7\text{-}y_4 & y_6\text{-}y_4 & y_5\text{-}y_4 & \end{array} \cdot \left(\begin{array}{ccccc} y_1\text{-}y_9 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & \\ 1 & 1 & y_3\text{-}y_7 & & \\ 1 & & & & \end{array} + \begin{array}{ccccc} y_1\text{-}y_9 & y_1\text{-}y_8 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & \\ 1 & 1 & 1 & & \\ 1 & & & & \end{array} + \begin{array}{ccccc} y_1\text{-}y_9 & 1 & 1 & 1 & 1 \\ y_2\text{-}y_9 & 1 & 1 & 1 & \\ 1 & 1 & 1 & & \\ 1 & & & & \end{array} \right)$$

Idea of the Proof

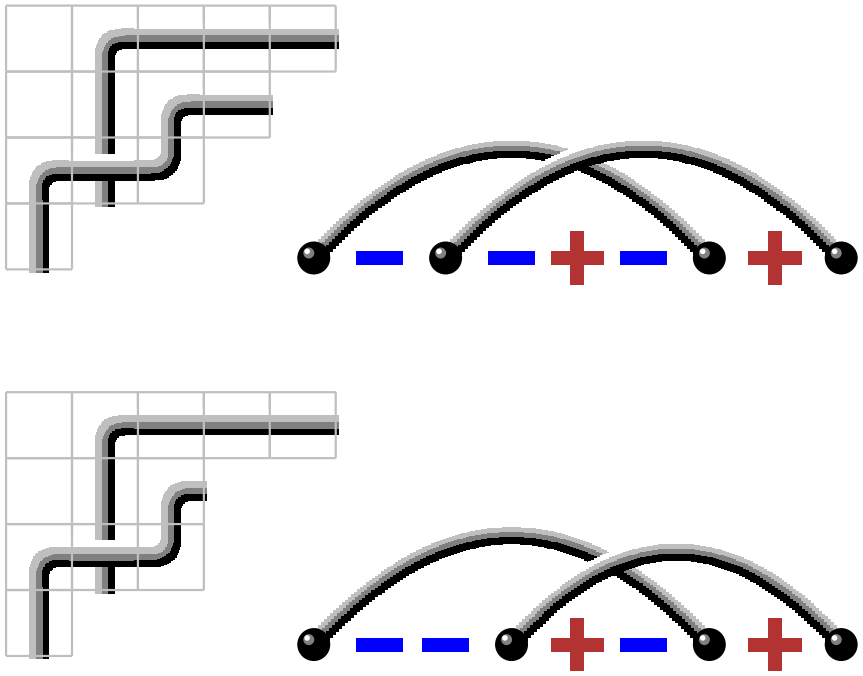
Geometric part of the proof

The initial case is by study the local structure of closed K-orbits at id.



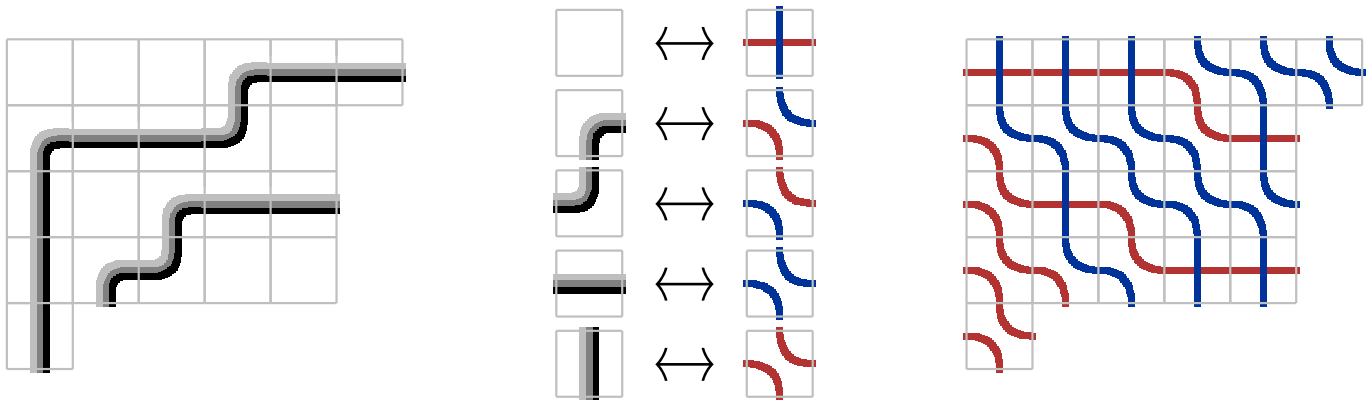
Algebraic part of the proof

The inductive step is by study the local change of a clan.



Relation to Billey formula

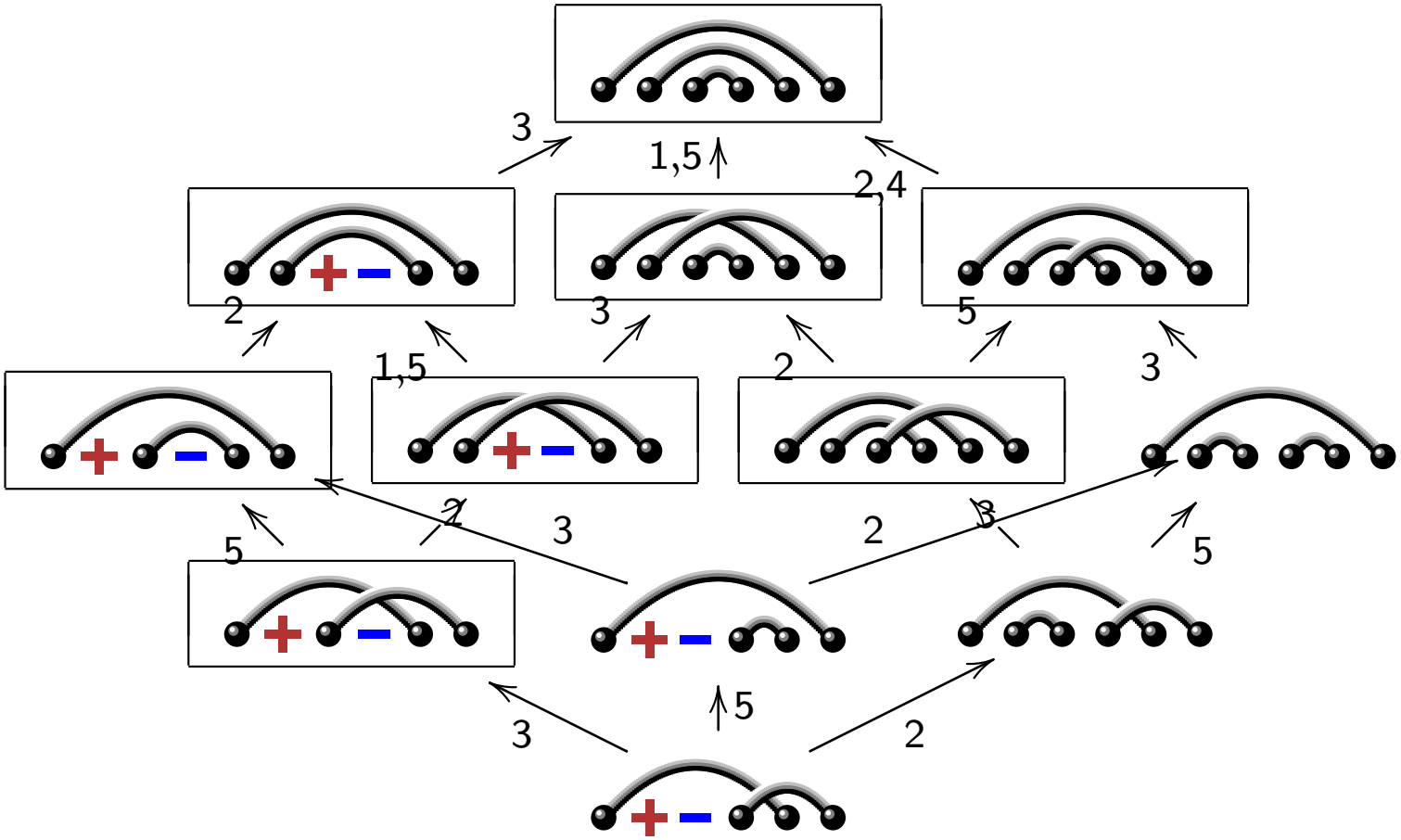
When γ is matchless, the K-orbit closure is a Richardson variety. In this case, $\Upsilon_\gamma|_{v_\gamma} = \mathfrak{S}_{v_\gamma}(v_\gamma y, y).\mathfrak{S}_{u_\gamma}(v_\gamma y, \overleftarrow{y})$. After some work, one can see our formula agrees with Billey formula [1]. Actually, the key step is a classical bijection of five-vertex model:



Note that does not appear since γ is matchless.

S. C. Billey. Kostant polynomials and the cohomology ring for G/B. Duke Math. J., 96(1):205–224, 1999.

Applications



Application — Schubert expansion

We give a complete answer of [WY14, Question 2]. Assume

$$\Upsilon_\gamma(x;y) = \sum_{w \in S_n} c_{\gamma w}(y) \mathfrak{S}_w(x;y).$$

Theorem (Chen–Fan–X.–Yao, 2025+)

We have

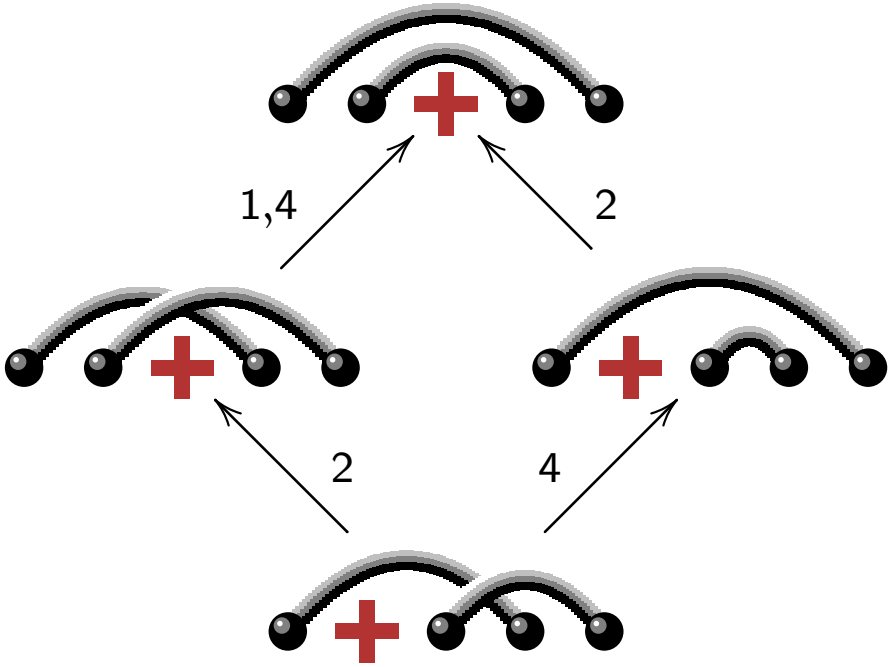
$$c_{\gamma w}(y) = \begin{matrix} \textit{Schubert polynomial} \\ \textit{of a partial permutation} \end{matrix} \textit{ or } 0$$

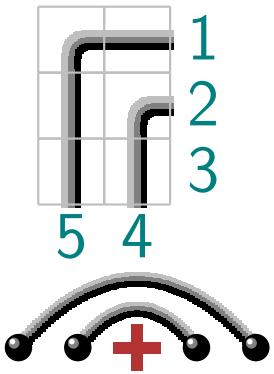
where the partial permutation is determined by $w * \gamma$.

Example

$\Upsilon_{\gamma=}$ (x,y)

$$= (y_1 - y_5)(y_2 - y_5)\mathfrak{S}_{\mathrm{id}}(x,y)$$
$$+ (y_1 - y_5)\mathfrak{S}_{s_2}(x,y)$$
$$+ (y_2 - y_4 + y_1 - y_5)\mathfrak{S}_{s_4}(x,y)$$
$$+ \mathfrak{S}_{s_1 s_2}(x,y)$$
$$+ \mathfrak{S}_{s_2 s_4 = s_4 s_2}(x,y).$$





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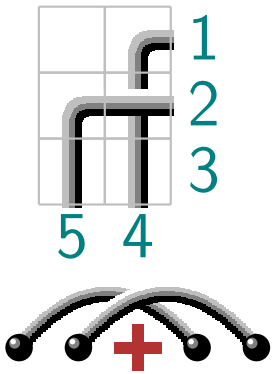
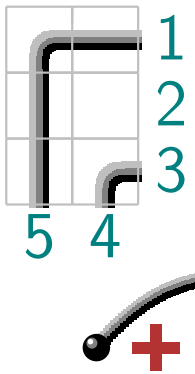
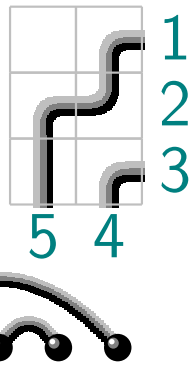
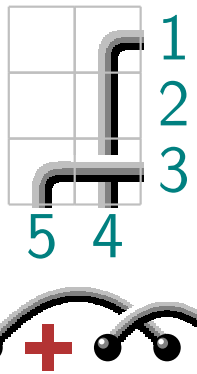


$y_1 - y_5$





$y_2 - y_4 + y_1 - y_5$

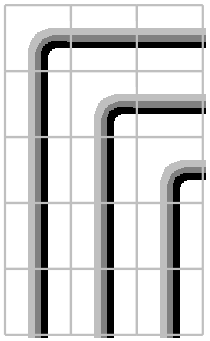


$(y_1 - y_5)(y_2 - y_5)$

Non-equivariant expansion

A general theorem of Brion [1] shows that the K-orbit closure is a multiplicity-free variety, i.e. the the coefficient of the non-equivariant Schubert expansion $c_{\gamma,w}(0) \in \{0,1\}$.

In our case, this can be seen combinatorially:



In general, it is not clear what is the equivariant analogue of "multiplicity-free variety".

M. Brion. On orbit closures of spherical subgroups in flag varieties. Comment. Math. Helv. 76(2) 2001, 263–299.

Corollary — structure constants

Since when γ is non-crossing, the K-orbit closure is a Richardson variety, we can obtain the following result.

Corollary (Chen–Fan–X.–Yao, 2025+)

For any non-crossing clan γ and any $v \in S_n$, assume that

$$\mathfrak{S}_{v_\gamma}(x;y)\mathfrak{S}_v(x;y) = \sum \mathbf{d}^{w}_{u_\gamma,v}(y)\mathfrak{S}_w(x;y).$$

Then the coefficient of $\mathfrak{S}_{w_0 v_\gamma}(x;y)$ is given by

$$\mathbf{d}^{w_0 v_\gamma}_{u_\gamma,v}(y) = c_{\gamma,w_0 v}(\overleftarrow{y}).$$

Note that this is a formula of Graham-positivity [Gr].

William Graham. Positivity in equivariant Schubert calculus. Duke Math. J., 109(3):599–614, 2001.

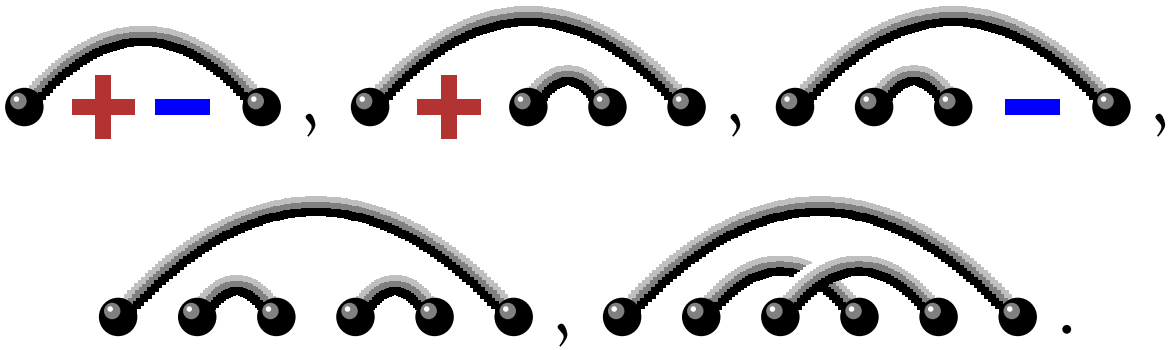
Application Smoothness

Localization can be used to detect smoothness; see [1].

Corollary (Chen–Fan–X.–Yao, 2025+)

For a clan γ, the following statements are equivalent:

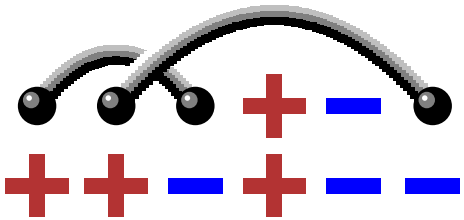
- *the orbit closure Y_γ is smooth at v_γ;*
- *the clan γ has exactly one BPD fragment;*
- *the clan γ avoids the following 5 patterns:*

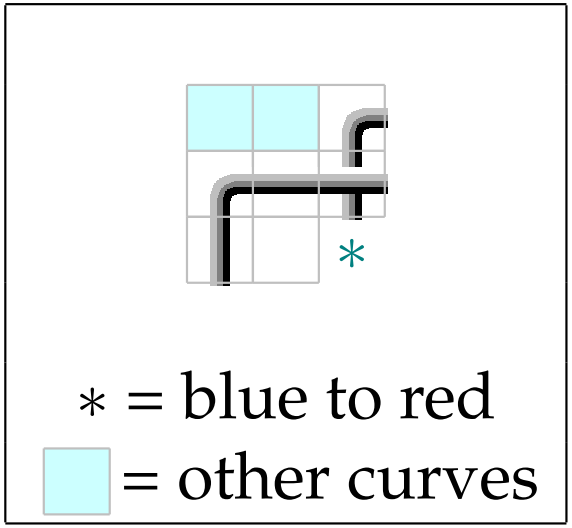


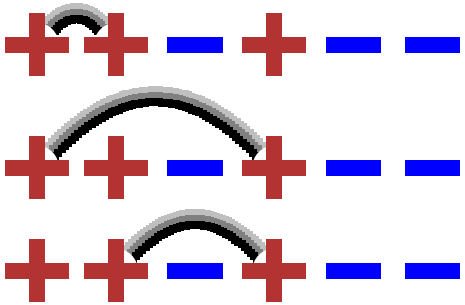
M. Brion. Equivariant cohomology and equivariant intersection theory. Representation Theories and Algebraic Geometry. vol 514 1998, 1–37.

Description of T-stable curves

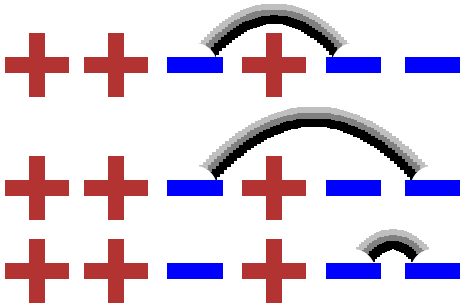
To apply the smooth criterion, we need to describe the T-stable curves containing v_γ in the K-orbit closure. There are three types:



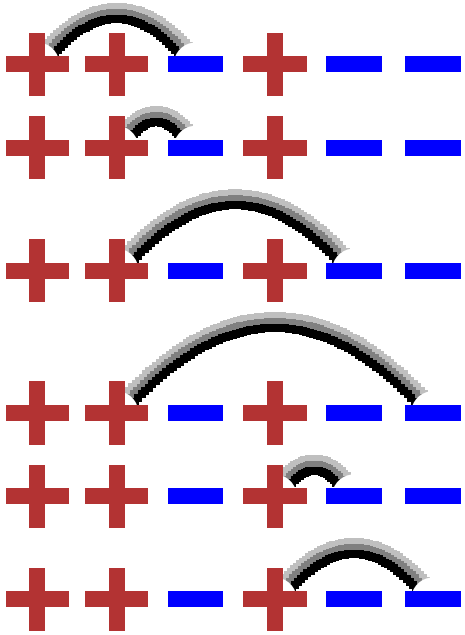




red to red



blue to blue



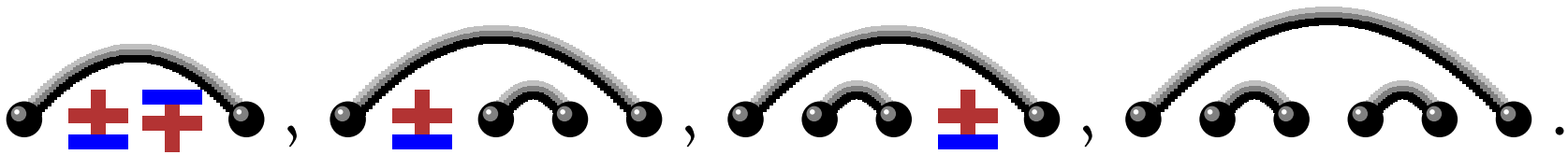
red to blue & under a bow

Corollary — smoothness

We can recover the smoothness criterion of McGovern [1] and Woo and Wyser [2].

Theorem (McGovern, Woo–Wyser)

For a non-crossing γ, the orbit closure Y_γ is smooth if and only if γ avoids the following patterns:



W. McGovern, Closures of K-orbits in the flag variety for $U(p, q)$, J. Alg. 322 (2009), 2709–2712.

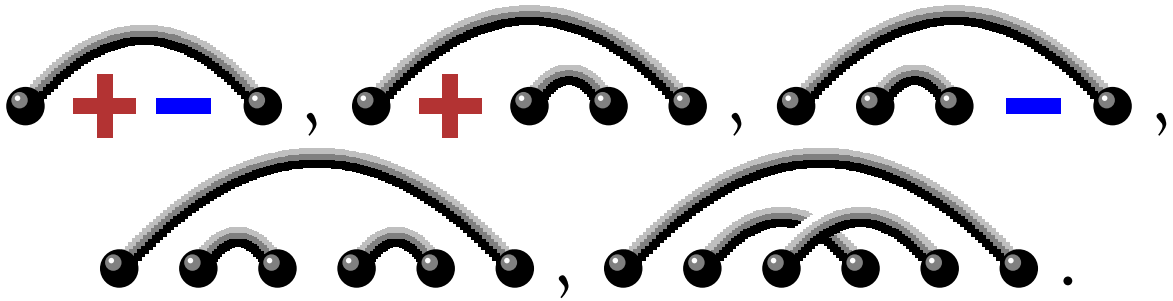
A. Woo and B. J. Wyser. Combinatorial results on (1, 2, 1, 2)-avoiding $GL(p, C) \times GL(q, C)$-orbit closures on $GL(p + q, C)/B$, Int. Math. Research Notices 24 (2015), 13148-13193.

The proof

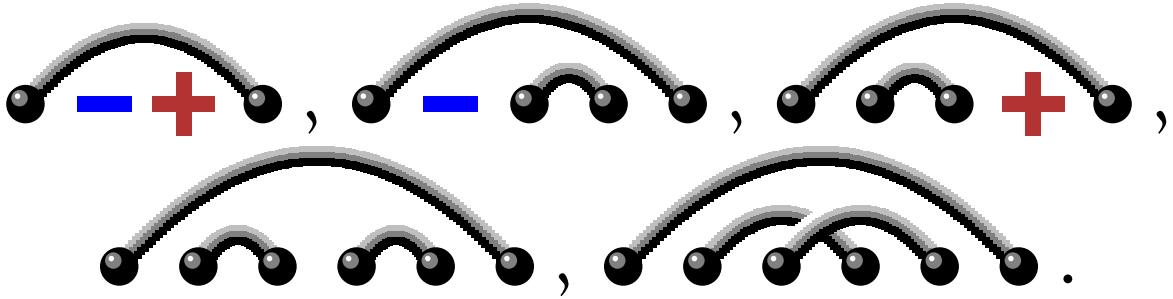
By [1], a Richardson variety is smooth if and only if it is smooth at the maximal and the minimal permutations.

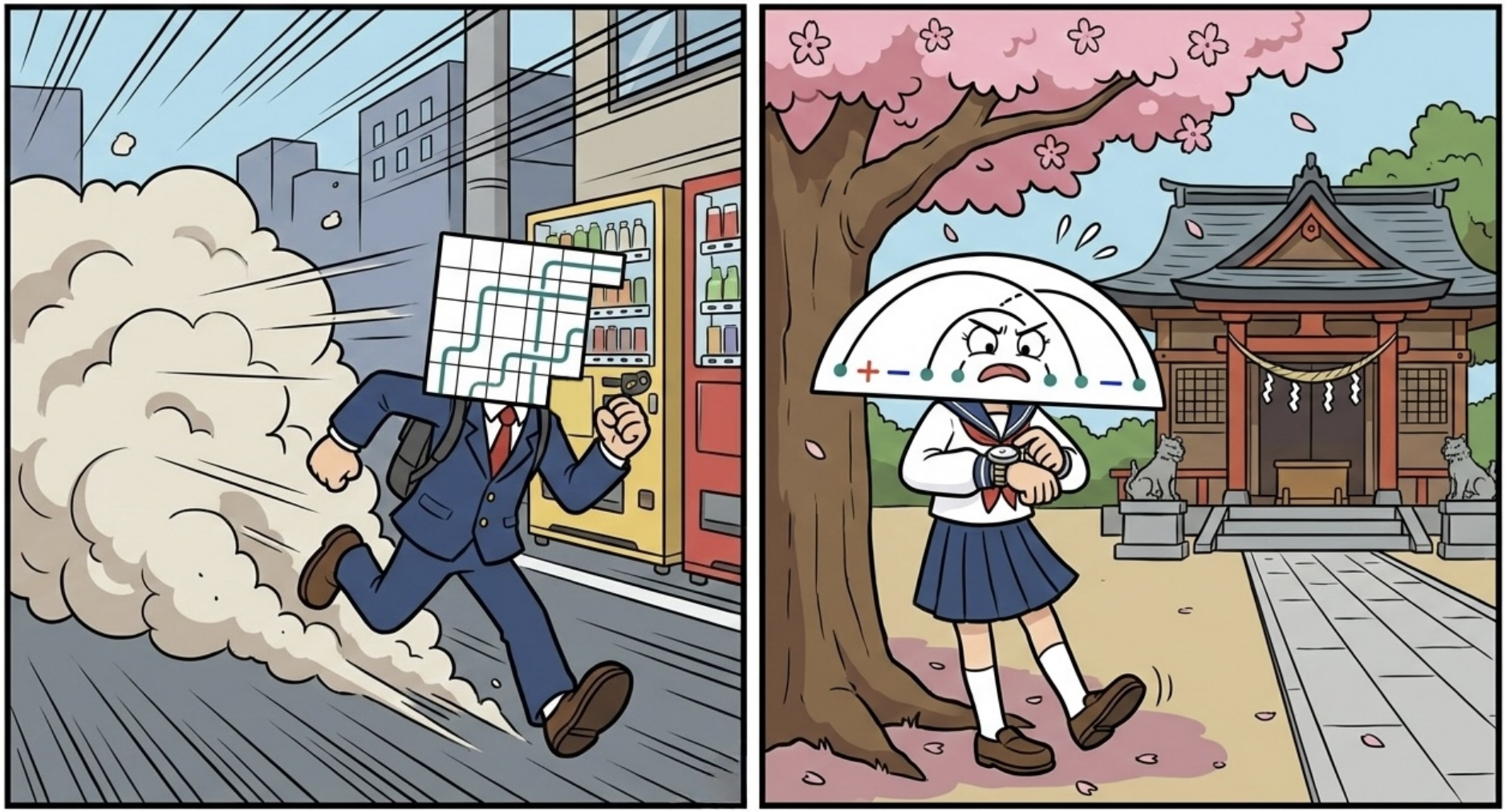
Richardson

smooth at v_γ
(minimal point)



smooth at $w_0 u_\gamma$
(maximal point)



A. Knutson, A. Woo, and A. Yong. Singularities of Richardson varieties. Mathematical Research Letters.

Thank You!